

Team 201 – Future Energy Needs at a Glance

Motivation/Introduction

- Current energy forecasts in the US focus on short-term or annual changes to energy demand, with long term predictions looking at a specific characteristic of energy production and demand. There currently exists no long-term predictive model of US energy demand that can help guide long-term decisions on both the policy or housing market level.
- As energy demand rises, adaptive load forecasting is critical for energy security, economic stability, and sustainability. Effective forecasting enables grid operators and policymakers to anticipate fluctuations in energy consumption, optimize resource allocation, and integrate renewable energy sources more efficiently.

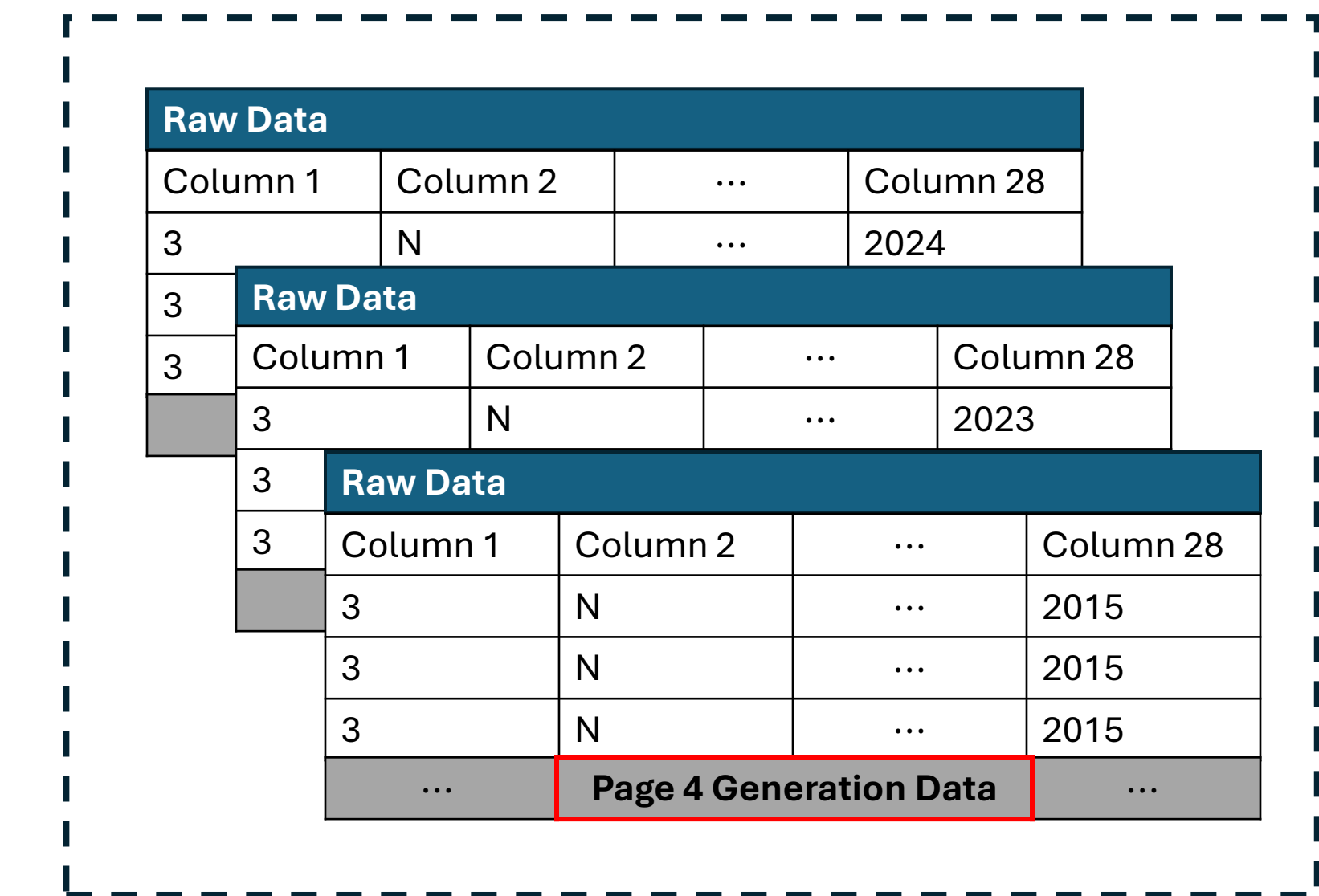
Approach

- Time-series Modeling
 - ARIMA: Regresses based on the data being stable and stationary, while also evaluating errors
- Convolutional Neural Network (CNN) – Long Short-Term Memory (LSTM) Hybrid Architecture
 - Combines spatial and temporal feature extraction in time-series data to provide predictions, in this case, by taking the previous months and predicting the next month
- Choropleth Map
 - Provides distinction and focus for future investments on a state-by-state basis.

Data

Raw Dataset

- The raw dataset, *Form EIA-923*, was obtained from the U.S. Energy Information Administration, including both independent data and analysis files covering the period from 2014 to 2024.
- Contains monthly and annual plant- and prime-mover-level data on electricity generation, fuel consumption, fossil-fuel stocks, and receipts (≈3,034 plants in the monthly survey; ≈12,562 in the annual release)
- Size: ~180 MB total (~18 MB per year over 10 years)



Data Preprocessing

- *Data_Clean_Code.py* extracts and consolidates each year’s “Page 4 Generation Data” tab from the raw files, producing a unified cleaned dataset for analysis
- Any missing values are filled to ensure a complete, analysis-ready dataset



Cleaned Dataset

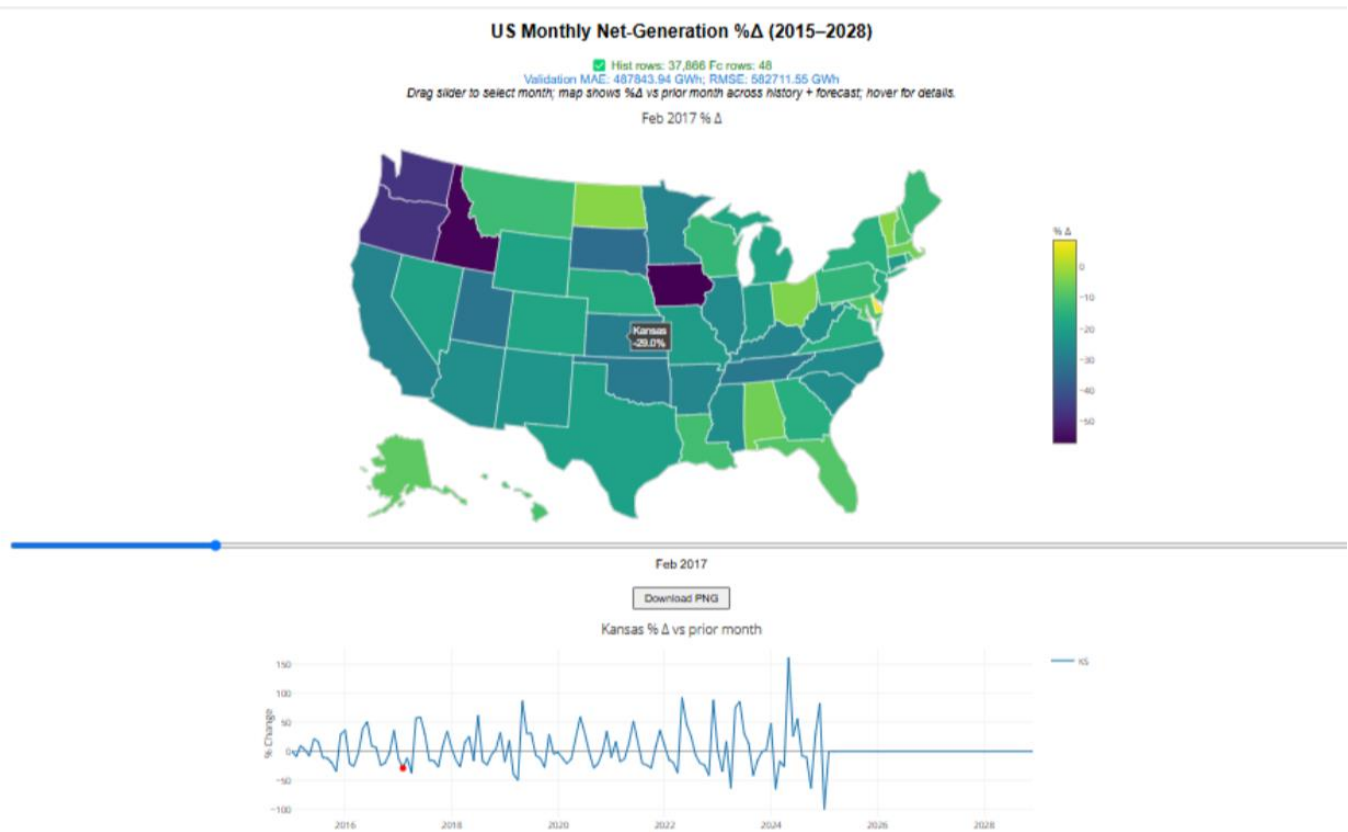
- Includes monthly net generation (MWh) for each year from 2015-2024
- Size: ~8 MB total; ~4,000 records per year across 28 fields

Cleaned Data			
Column 1	Column 2	...	Column 28
3	N	...	2024
3	N	...	2024
3	N	...	2024
3	N	...	2023
3	N	...	2023
3	N	...	2023
⋮	⋮	⋮	⋮
3	N	...	2015
3	N	...	2015
3	N	...	2015

Experiments and Results

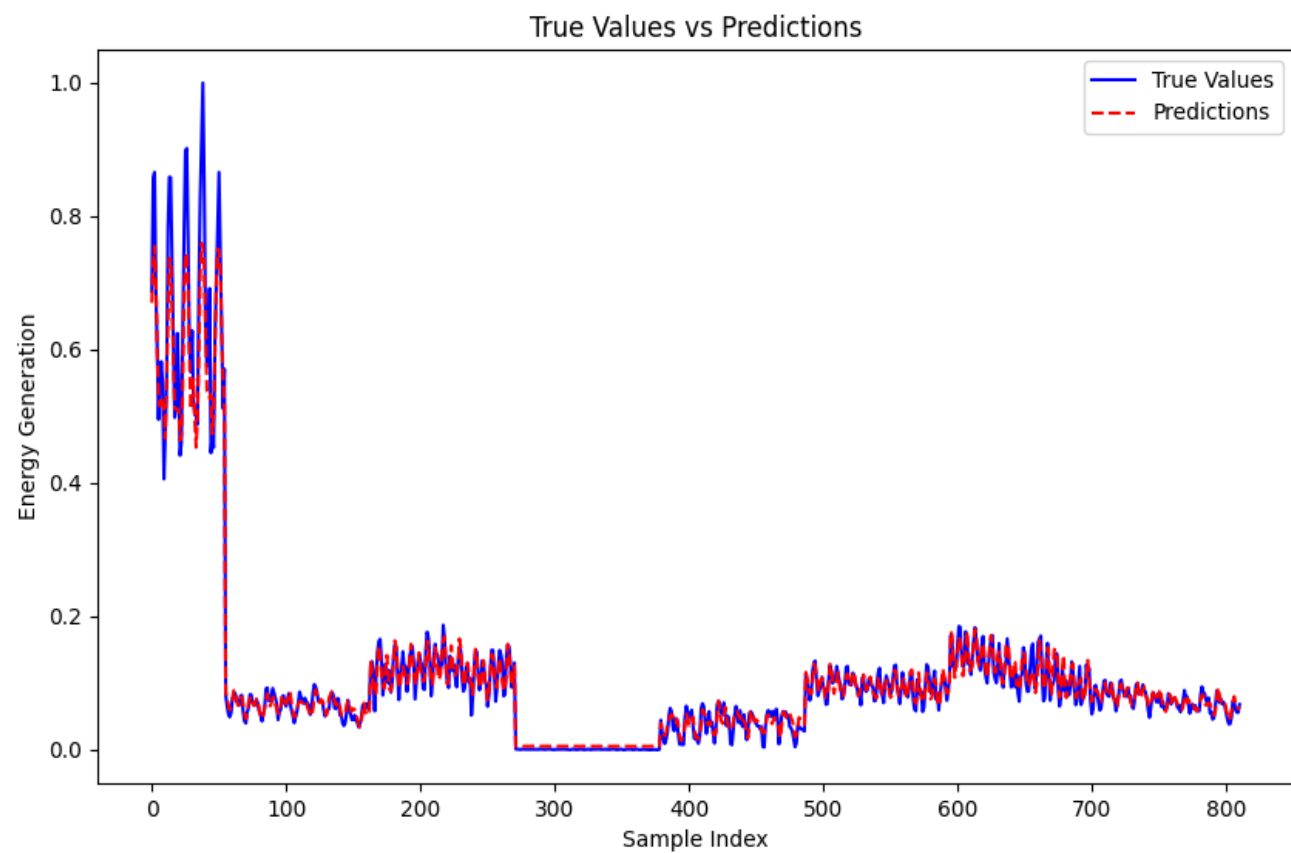
Visualization

- Deployed on a local HTTP server to measure lag time



CNN-LSTM

- Forecasts on the held-out test set were assessed using: MSE, MAE, RMSE, and R²

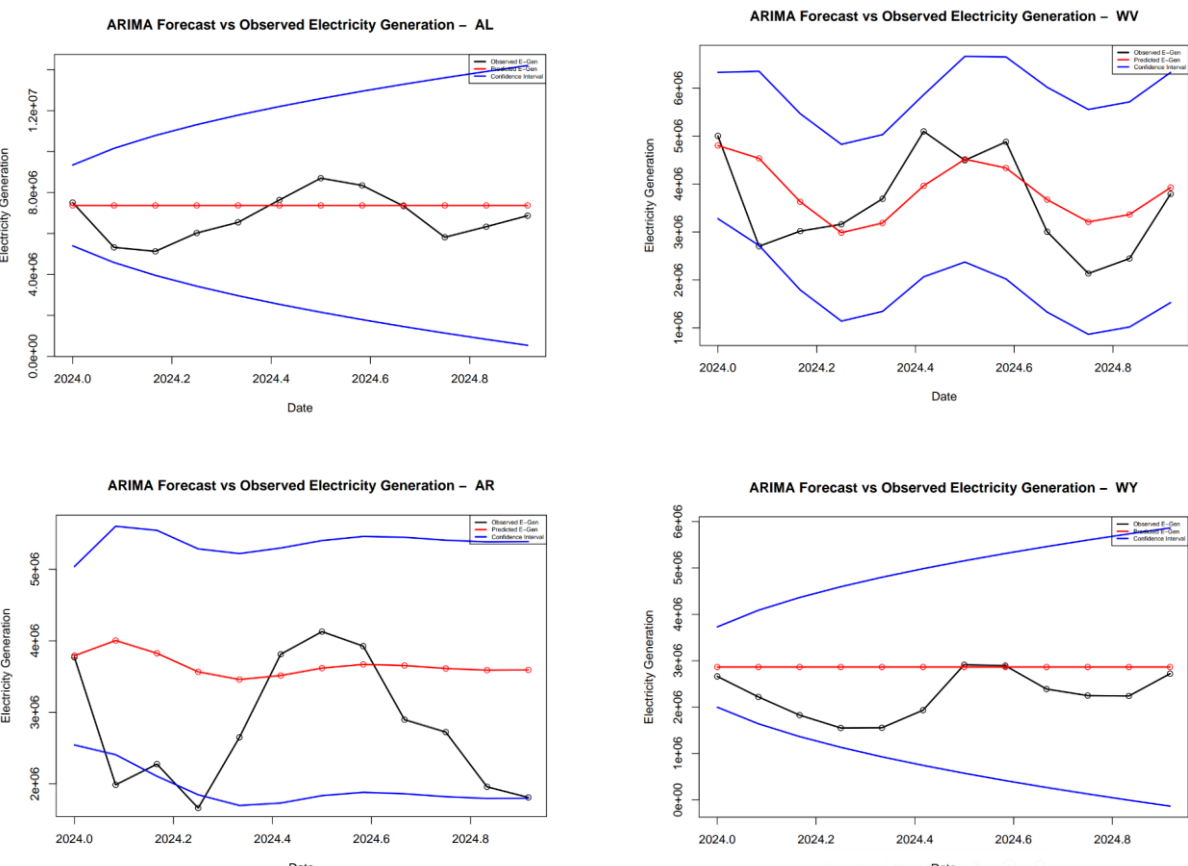


state_forecasts_5yr_monthly				
Plant State	Forecast Date	Forecasted Net Generation	Month	Year
AK	1/1/2025	288358.9	January	2025
AK	2/1/2025	318504	February	2025
AK	3/1/2025	316675.7	March	2025
⋮	⋮	⋮	⋮	⋮
WY	11/1/2029	2469535	November	2029
WY	12/1/2029	2889275	December	2029

Test MSE (loss)	Test MAE	RMSE	R ²
0.0006231	0.01275	0.02496	0.9743

ARIMA

- Twelve-month-ahead forecasts for 2024 were generated from an ARIMA model trained on 2015-2023 data, then compared to the actual 2024 monthly values using MAPE, PM, RMSE, MAE, and ME.



State	MAPE	PM	RMSE	MAE	ME
AK	3.5876	922.9638	6428257	253413.1	-253413
AL	0.167039	1.274194	2253759	1025490	-572671
AR	0.485797	2.051764	3421179	1035638	-858582
AZ	0.193814	0.640284	2294868	923818.1	-32337.8
CA	0.322187	0.843823	1877239	1551659	-690499
CO	0.21817	1.114425	4010815	472582.2	-352024
⋮	⋮	⋮	⋮	⋮	⋮
WY	0.327981	2.726975	4102334	614247.8	-601296

	MAPE	PM	RMSE	MAE	ME
Average	1.175	20.29	4646541	713771.2	-273577